BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SWAT

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2023

BIOLOGY

Past Paper 2023 • Paper Code: 102301

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 12

Note: 1) Attempting all MCQs is compulsory. This paper along with the OMR sheet must be returned to the superintendent after due time. 2) Fill the circle A B C D, which one is correct with blue or black ball point in separate OMR Sheet like ● 3) If more than one circle in the OMR sheet is filled then no credit will be given to such answer.

Q1. Oxygen from the lungs enter into the blood due to

(A) Diffusion
(B) Less affinity of O_2 for air.
(C) Higher affinity of CO_2 for air
(D) Osmosis

Q2. The loss of water from the plants in the form of small droplets is _____.

(A) Stomatal transpiration
(B) Cuticular transpiration
(C) Lenticular transpiration
(D) Guttation

Q3. _____carry impulses from sense organs to the CNS.

(A) Sensory neurons
(B) Motor neurons
(C) Associative neurons
(D) Cerebrum

Q4. One of the following is a muscular layer of the eye ball :

(A) Retina
(B) Sclera
(C) Cornea
(D) Cone

Q5. The main protein in the matrix of Cartilage is_____

(A) Collagen
(B) Osteo nectin
(C) Keratin
(D) Actin

Q6. Muscles are attached to bones by.

(A) Ligaments
(B) Tendons
(C) Cartilage
(D) None of these

Q7. In a flower embryo sac is formed inside the _____.

(A) Anther
(B) Filament
(C) Ovule
(D) Style

Q8. All multicellular animals develop from a single cell , called :

(A) Egg
(B) Zygot
(C) Sperm
(D) Blood cells

Q9. Phenoty is the expression of _____.

(A) Genetic make up
(B) Physical make up
(C) Chemical composition
(D) Metabolism

Q10. An Ecological pyramid refers to the _____

(A) Pyramid of Energy
(B) Pyramid of number
(C) Pyramid of Biomass
(D) All of these

Q11. In a technique for genetic engineering ,rDNA Stands for :

(A) Genetic make up
(B) Replicative DNA
(C) Recombinant DNA
(D) Retro DNA

Q12. Vaccines can be administered through _____

(A) Injections
(B) Mouth
(C) Nose
(D) All of these

SECTION "B"

Time Allowed: 2 hours and 40 minutes | **Maximum Marks:** 32

QII. Attempt any EIGHT Parts out of the following. Each Part carries equal marks.

(i) Discuss gaseous exchange in lungs.

(ii) Describe Urinary System of Humans.

(iii) Write down the symptoms and treatment of kidney stone.

(iv) Define Neuron. Draw its structure.

(v) Differentiate between the two types of coordination system..

(vi) Write down names of vertebras according to their location in the body.

(vii) Write down conditions necessary for Germination of seeds.

(viii) Differentiate between continuous and Discontinuous variation.

(ix) Define Nitrogen cycle. Write down the steps involved in nitrogen cycle.

(x) Write down major uses of single cell protein.

(xi) Give two examples of Drugs which are extracted from plants.

SECTION "C"

Maximum Marks: 21

Note: Attempt any THREE questions of the following. Each question carries equal Marks.

QIII.

(a) [4] Explain the effect of exercise on the rate of breathing .

(b) [3] Describe Rheumatoid arthritis under the following headings. i) Causes ii) Symptoms iii) Treatment

QIV.

(a) [4] Define self pollination and Cross pollination and give examples..

(b) [3] Write down Mode of Transmission of AIDS.

QV.

(a) [3] Explain why Mendel selected peas as a subject for studying genetics.

(b) [4] Differentiate between Population and community.

QVI.

(a) [4] Define Fermentation. Discuss the methods of fermentation.

(b) [3] Describe sulfonamides and Tetracyclines.